Carolyn B. Voter, PhD

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RESEARCH INTERESTS

- Hydrologic feedbacks: surface water-groundwater interactions, land-atmosphere interactions
- Integrated hydrologic modeling, high throughput/performance computing
- Urban ecohydrology, green stormwater infrastructure, stormwater management
- Sustainable and resilient communities, coupled human-natural systems

EDUCATION

2019	**Ph.D.**	*University of Wisconsin-Madison*	Civil Engineering *Minor:* Subsurface Hydrologic Processes *Thesis:* Hydroecologic effects of urban development decisions in residential areas *Advisor:* Dr. Steven P. Loheide II
2012	**B.S.**	*Bucknell University*	Civil Engineering *summa cum laude*

PROFESSIONAL APPOINTMENTS

2026–present	*Assistant Professor*	Dept. of Civil and Environmental Engineering, UW-Madison
2022 – 2026 **2025 – 2026*	*Assistant Professor*	**Brigid Hawk McVaugh '74 and John B. McVaugh Jr. '71 '75M Term Professor of Environmental Engineering*,** Dept. of Civil, Construction, and Environmental Engineering, Dept. of Earth Sciences, University of Delaware
2019 – 2021	*Postdoctoral Fellow*	**Wisconsin Water Resources Science-Policy Fellowship,** Wisconsin Water Resources Institute, UW-Madison
2018 – 2019	*Graduate PA*	**Water@UW-Madison Project Assistant**, Aquatic Sciences Center, UW-Madison
2015 – 2016	*Graduate Fellow*	**Wisconsin Water Resources Science-Policy Fellowship,** Wisconsin Water Resources Institute, UW-Madison
2014 – 2015	*Graduate Fellow*	**Weston Fellowship,** Dept. of Civil and Environmental Engineering, UW-Madison
2016 – 2018, 2013 – 2014	*Graduate RA*	**Research Assistant**, Dept. of Civil and Environmental Engineering, UW-Madison
2012	*Graduate TA*	**Teaching Assistant**, CEE 311: Hydroscience, UW-Madison
2008 – 2012	*Undergrad. Fellow*	**Presidential Fellowship,** Bucknell University

AWARDS AND HONORS

As an *Assistant Professor*

2025 – 2026 **Brigid Hawk McVaugh ’74 and John B. McVaugh Jr. ‘71’75M Term Professor of Environmental Engineering**, University of Delaware

2025 **Most Valuable Professor**, Women’s Lacrosse Team, University of Delaware
Selected by a member of the UD Women’s Lacrosse Team as their Most Valuable Professor.

2024 **UD Excellence in Teaching Award**, University of Delaware
Awarded annually via student nominations and committee selection to up to 4 faculty (out of ~1,300) who are intellectually demanding and rigorous, deeply committed to teaching and students, and have a positive and lasting impact.

2022 **Most Valuable Professor**, Women’s Volleyball Team, University of Delaware
Selected by a member of the UD Women’s Volleyball Team as their Most Valuable Professor.

As a *Postdoctoral Fellow*

2021 **Team of the Year**, Environmental Management Division, Wisconsin Department of Natural Resources
Awarded to members of the Central Sands Lake Study Team.

2021 **AGU 2020 Editor’s Citation for Excellence in Refereeing**, Water Resources Research

As a *Graduate Student*

2019 **James R. Villemonte Excellence in Research Award**, Dept. of Civil and Environmental Engineering, UW-Madison
Awarded annually to a Hydrology, Fluid Mechanics, and Water Resources Engineering graduate student in recognition of their scientific contributions.

2019 **Outstanding Graduate Oral Presentation**, American Water Resources Association – Wisconsin Section Annual Meeting

2018 **Outstanding Student Presentation Award**, American Geophysical Union Fall Meeting

2012 **Chancellor’s Opportunity Award**, UW-Madison

As an *Undergraduate Student*

2012 **Peer Award of Distinction**, Dept. of Civil Engineering, Bucknell University
Recognizes the student, selected by the student's peers, who has been most generous and unselfish in assisting and working with his or her classmates during their education at Bucknell University, and who exemplifies the qualities of the Civil and Environmental Engineering profession including integrity, honor, and service.

2012 **Michael D. LaGrega Award for Excellence in Environmental Engineering**, Bucknell University
Awarded to a member of the graduating class in civil and environmental engineering whose academic achievements and interests show outstanding promise for a career in environmental engineering.

2012 **Eleanor D. Decker Prize for Women**, Bucknell University
Awarded to the woman of the graduating class who best represents the ideals fostered by Bucknell University: intellectual maturity, personal conviction and strength of character.

2011 **Chi Epsilon**, Civil Engineering Honor Society

2011 **Tau Beta Pi**, Engineering Honor Society

GRANTS

Active

2025 - 2029	*Senior Personnel*	**E-RISE RII: An Investigation of Saltwater Intrusion in the Complex Coastal Environment of Delaware** Funding: National Science Foundation, EPSCoR Research Incubators for STEM Excellence Collaborators: Kent Messer (Lead PI, UD) + 4 Co-PIs + 8 Senior Personnel (UD, Delaware State University, Delaware Technical Community College, Goldey-Beacom College, Wilmington University)	$7,000,000
2024 – 2027	*Lead PI*	**SAI: Examining Long-Term "Success" of Green Stormwater Infrastructure.** Funding: National Science Foundation, Strengthening American Infrastructure Program. Collaborators: Paul Imhoff (UD), Rebecca Nixon (University of Washington), Yao Hu (UD)	$749,964
2024 – 2027	*Lead PI*	**Integrated Assessment of Climate Change Impacts to Groundwater, Stormwater, and Wastewater Infrastructure at Coastal Military Facilities** Funding: Department of Defense, Environmental Security Technology Certification Program Collaborators: Paul Imhoff (UD), Holly Michael (UD), Larry Trout (Straughan Environmental)	$1,541,829

Completed

2024 – 2026	*Co-PI*	**An Investigation of Green Infrastructure to Enhance Resilient Stormwater Management in the City of Wilmington** Funding: Delaware Sea Grant Collaborators: Vaishnavi Tripuraneni (UD), Victor Perez (UD), Rebecca Nixon (University of Washington)	$275,013
2022 – 2024	*Lead PI*	**SAI-P: Optimizing Deployment of Green Stormwater Infrastructure for Maximum Benefit** Funding: National Science Foundation, Strengthening American Infrastructure Program. Collaborators: Paul Imhoff (UD), Rebecca Nixon (UD), Yao Hu (UD)	$123,556
2020 – 2022	*Associate Investigator*	**Mitigating Fooding, Extreme Heat, and Heavy Rainfall in Urban Coastal Communities by "Greening"** Funding: Wisconsin Sea Grant Collaborators: Steve Loheide (Lead PI, UW-Madison), Dan Wright (UW-Madison)	$238,582

PUBLICATIONS

[us] = undergraduate student, [gs] = graduate student, [m] = mentor

Peer-Reviewed Papers

8. Alexander, G.A. [gs], D.B. Wright, **C.B. Voter**, S.P. Loheide II [m]. 2026. Ecohydrologic Processes Modify Urban Rainfall Intensification in Land-Atmosphere Simulations. *Geophysical Research Letters, 53,* e2026GL121642. https://doi.org/10.1029/2026GL121642

7. Alexander, G.A. [gs], D.B. Wright, **C.B. Voter**, S.P. Loheide II [m]. 2025. City-Scale Evaluation of Urban Ecohydrologic Processes on Surface Energy Balance, Heat, and Humidity. *Urban Climate,* 63, 102596. https://doi.org/10.1016/j.uclim.2025.102596

6. Alexander, G.A. [gs], **C.B. Voter**, D.B. Wright, S.P. Loheide II [m]. 2024. Urban Ecohydrology: Resolving Sub-Grid Surface Lateral Water and Energy Transfers in a Land Surface Model. *Water Resources Research.* https://doi.org/10.1029/2023WR035511

5. Oswald, C.J., C. Kelleher, S.H. Ledford, K.G. Hopkins, A. Sytsma, D. Tetzlaff, L. Toran, **C.B. Voter.** 2023. Moving beyond impervious surface cover: the need for integrated process understanding of water flow in cities. *Journal of Hydrology.* https://doi.org/10.1016/j.jhydrol.2023.129188

4. **Voter, C.B**., F.J. Guerrero-Bolaño, A.W. Latzka, B.M. Maitland, and J. Hauxwell. 2021. Adaptable University-Agency Early-Career Fellowship Program Creates a Win-Win-Win for Wisconsin's Waters. *Journal of Contemporary Water Research & Education,* 174:139-154. https://doi.org/10.1111/j.1936-704X.2021.3365.x

3. **Voter, C.B.** and S.P. Loheide II [m]. 2021. Climatic controls on the hydrologic effects of urban low impact development practices. *Environmental Research Letters,* 16(6): 064021. https://iopscience.iop.org/article/10.1088/1748-9326/abfc06

2. **Voter, C.B.** and S.P. Loheide II [m]. 2020. Where and When Soil Amendment is Most Effective as a Low Impact Development Practice in Residential Areas. *Journal of the American Water Resources Association,* 56(5):776-789. https://doi.org/10.1111/1752-1688.12870

1. **Voter, C.B.** and S.P. Loheide II [m]. 2018. Urban Residential Surface and Subsurface Hydrology: Synergistic Effects of Low-Impact Features at the Parcel-Scale. *Water Resources Research, 54*. https://doi.org/10.1029/2018WR022534

Refereed Conference Papers

1. Mayo, L. and **C.B. Voter.** 2019. Introducing Students to Engineering by Helping Them RePicture Their World. ASEE Annual Conference & Exposition, Tampa, FL. https://peer.asee.org/32265. (17p.)

Technical Reports

3. **Voter, C.B.**, R. Nixon, P. Imhoff, A. Slocum, R. Zobel[gs], and Y. Hu. 2024. Effective and Equitable Green Stormwater Infrastructure in the Chesapeake Bay: Current Challenges and Next Steps. University of Delaware.

2. **Voter, C.B.**, C. Hein, J. Chenevert, I. Anderson, R. Smail, M. Gibson, K. Doyle, S. Bunde. 2021. Appendix B: Central Sands Lakes Study Technical Report: Lake Ecosystem Characterization and Response. Wisconsin Department of Natural Resources. https://doi.org/10.5281/zenodo.5708751

1. **Voter, C.B.** and E. Verbeten. February 2017. “Groundwater: Powering Wisconsin’s Economy.” *Wisconsin Natural Resources*. Wisconsin Department of Natural Resources. http://dnr.wi.gov/wnrmag/2017/02/Insert1.pdf

PRESENTATIONS

[us] = undergraduate student, [gs] = graduate student, [m] = mentor

Conference Abstracts (selected)

2026 Erukubami, O.[gs], A. Farnum [s], D. Pratt, H. Michael, P. Imhoff, L. Trout, T. Barber, C. Overcash, **C.B. Voter**. Examining the Impacts of Bidirectional Interactions Between Surface and Subsurface Systems in a Coupled Coastal Urban Hydrologic Model. Groundwater Modeling and More. Princeton, NJ. Jun 2026. *Poster*

2025 Farnum, A.[gs], O. Erukubami [gs], **C.B. Voter,** H. Michael, P. Imhoff, T. Barber, L. Trout, and C. Overcash. Representing Fast (Surface) and Slow (Subsurface) Threats to Below-Ground Water Infrastructure in Integrated Urban Coastal Hydrologic Modeling. American Geophysical Union Fall Meeting. New Orleans, LA. Dec 2025. *Poster*.

Alexander, G.A.[gs], E. Dougherty, D.B. Wright, **C.B. Voter**, S.P. Loheide II[m], C. Vanorio, M. Ozdogan, A. Schneider. Modeling Future Heat Extremes in Milwaukee Wisconsin Using Extremes-Preserving Psuedo-Global Warming Methods. American Geophysical Union Fall Meeting. New Orleans, LA. Dec 2025. *Oral.*

Amiri, Z., A. Bhaskar, and **C.B. Voter.** Evaluating the Effects of Varying Urban Irrigation Levels on Groundwater Recharge and Surface Runoff at the Parcel Scale. American Geophysical Union Fall Meeting. New Orleans, LA. Dec 2025. *Poster*.

Alexander, G.A.[gs], **C.B. Voter**, D.B. Wright, and S.P. Loheide II[m]. Impacts of Nature-Based Solutions on Hydrology and Extreme Heat within Coupled Land-Atmospheric Simulations. American Geophysical Union Fall Meeting. New Orleans, LA. Dec 2025. *Oral.*

Zobel, R.[gs], E. Rodden, V. Tripuraneni, V. Perez, and **C.B. Voter**. Hydrologic Modeling Meets Community Voices: Advancing GSI in Northeast Wilmington. American Geophysical Union Fall Meeting. New Orleans, LA. Dec 2025. *Poster*.

Erukubami, O.[gs], **C.B. Voter**. Modeling Coupled Stormwater-Wastewater-Groundwater Dynamics in Urban Coastal Settings. Geological Society of America. San Antonio, TX. Oct 2025. *Poster*.

Alexander, G.A.[gs], D.B. Wright, **C.B. Voter**, and S.P. Loheide II[m]. Quantifying the City-Scale Benefits of Urban Vegetation and Green Infrastructure using Novel Land-Atmosphere Simulations. European Geophysical Union General Assembly Conference. Apr 2025. *Oral.*

2024 Ruggiero, E.[gs] and **C.B. Voter**. Aging Green Infrastructure Still Effective?: Investigating Long-Term Research on the Eco-hydrologic Performance of Rain Gardens. American Geophysical Union Fall Meeting. Washington D.C. Dec 2024. *Oral*.

Farnum, A.[gs] and **C.B. Voter**. Converting Future Precipitation Estimates from Low to High Temporal Resolutions: A Workflow for the Surface Water Modeler. American Geophysical Union Fall Meeting. Washington D.C. Dec 2024. *Oral*.

Zobel, R.[gs] and **C.B. Voter**. From Isolated Struggles to Collaborative Futures: Investigating Stormwater Challenges and Opportunities Across University Campuses. American Geophysical Union Fall Meeting. Washington D.C. Dec 2024. *Poster*.

Erukubami, O.[gs], A. Farnum [gs], **C.B. Voter**. Integrated Modeling Assessment of Sea Level Rise Impacts on Groundwater and Water Infrastructure in an Urban Coastal Environment. American Geophysical Union Fall Meeting. Washington D.C. Dec 2024. *Poster*.

Alexander, G.A.[gs], D.B. Wright, **C.B. Voter**, and S.P. Loheide II[m]. Beyond Reduction of Runoff: Examining the Added Benefits of Widespread Urban Greening. American Geophysical Union Fall Meeting. Washington D.C. Dec 2024. *Poster.*

Erukubami, O.[gs], H. K. Morgan, L.G. Trout, Jr., **C.B. Voter**. Stormwater Management Facilities in Coastal Areas with Respect to Climate Change. Villanova University Stormwater Symposium 2024. Villanova, PA. Oct 2024. *Oral.*

Farnum, A.[gs] and **C.B. Voter**. Adapting Precipitation from Climate Models for Use in Stormwater Modeling. WaterSciCon24. St. Paul, MN. Jun 2024. *Poster.*

Erukubami, O.[gs], A. Farnum [gs], **C.B. Voter**. Integrated Modeling Assessment of Sea Level Rise Impacts on Groundwater and Water Infrastructure in an Urban Coastal Environment. WaterSciCon24. St. Paul, MN. Jun 2024. *Poster.*

Zobel, R.[gs] and **C.B. Voter**. Comprehensive Review of Stormwater Management Practices Across NIWR Member Institutions in the Mid-Atlantic Region. At:

- 2024 AWRA, UCOWR, NIWR Water Resources Conference. St. Louis, MO. Oct 2024. *Oral*.
- WaterSciCon24. St. Paul, MN. Jun 2024. *Poster.*

Erukubami, O.[gs], A. Farnum [gs], **C.B. Voter**. Assessment of Sea Level Rise Impacts on Shallow Groundwater in an Urban Coastal Environment. MODFLOW and More. Princeton, NJ. Jun 2024. *Poster.*

Zobel, R.[gs], **C.B. Voter**, R. Nixon, P. Imhoff, Y. Hu. Intersection of Maintenance, Effectiveness, and Equity in Green Stormwater Infrastructure in the Chesapeake Bay Region. ASCE EWRI Operations and Maintenance of Stormwater Systems Conference. Austin, TX. Apr 2024. *Oral.*

Alexander, G.A.[gs], **C.B. Voter**, S.P. Loheide II[m], and D.B. Wright. How Adoption of Green Infrastructure Impacts Urban Hydrologic-Atmospheric Processes. ASCE EWRI Environment & Water Resources Congress. Milwaukee, WI. May 2024. *Oral.*

Voter, C.B. Does This Matter? Lessons Learned (and Still Learning) from Connecting with Water Science Stakeholders. American Meterological Society Annual Meeting 23rd Annual Student Conference. Jan 2024. *Oral.*

>> Invited

2023 **Voter, C.B.**, P. Imhoff, H. Michael, L. Trout, O. Erukubami[gs], A. Farnum[gs]. Integrated Assessment of Climate Change Impacts to Groundwater, Stormwater, and Wastewater Infrastructure at Coastal Military Facilities. DoD Energy and Environment Innovation Symposium. Dec 2023. *Oral.*

>> Invited

Zobel, R.[gs], **C.B. Voter**. Identifying the Current State of Stormwater Management and Green Stormwater Infrastructure on Public University Campuses in the Chesapeake Bay Region. CUAHSI Biennial Meeting. Lake Tahoe, CA. Jun 2023. *Poster.*

Erukubami, O.[gs], **C.B. Voter**. Assessment of Sea Level Rise Impacts on Aquifer Systems in Coastal Environments: A Case Study of Bowers Beach, Delaware. CUAHSI Biennial Meeting. Lake Tahoe, CA. Jun 2023. *Poster.*

Alexander, G.A.[gs], **C.B. Voter**, S.P. Loheide II[m], and D.B. Wright. Better representation of urban hydrologic processes alters surface water and temperature cycles in regional coupled climate models. Consortium for the Advancement of Hydrologic Sciences, Inc. (CUAHSI) Biennial Meeting. Lake Tahoe, CA. Jun 2023. *Poster & Lightning Talk.*

Alexander, G.A.[gs], **C.B. Voter**, S.P. Loheide II[m], and D.B. Wright. Resolving Fine-Scale Lateral Water Transfers in Urban Environments Alters Regional Climate Simulations. American Meteorological Society Annual Meeting. Denver, CO. Jan 2023. *Poster.*

2022 Alexander, G.A.[gs], **C.B. Voter**, S.P. Loheide II[m], and D.B. Wright. Better Representation of Urban Hydrologic Processes Alters How Heat Responds to Urban Vegetation in Regional Climate Models. American Geophysical Union Fall Meeting. Chicago, IL. Dec 2022. *Poster.*

Voter, C.B., G.A. Alexcander[gs], S.P. Loheide II[m], D.B. Wright. Consider the Climate: Understanding What to Expect from Urban Hydrology Interventions Depending on Where You Are. American Geophysical Union Frontiers in Hydrology Meeting. San Juan, PR. Jun 2022. *Oral.*
>> Invited

2021 **Voter, C.B.**, C. Hein[m], J. Chenevert, I. Anderson. A Framework for Describing the Hydrologic Regime of Lakes for Improved Water Quantity Management. American Geophysical Union Fall Meeting. New Orleans, LA. Dec 2021. *Oral.*

Alexander, G.A.[gs], **C.B. Voter**, S.P. Loheide II[m], and D.B. Wright. Incorporating Impacts of Green Infrastructure into a Large-Scale Land Surface Model. American Geophysical Union Fall Meeting. New Orleans, LA. Dec 2021. *Oral.*

Alexander, G.A.[gs], **C.B. Voter**, S.P. Loheide II[m], and D.B. Wright. Incorporating the Hydrologic Impacts of Low Impact Development in a Large-Scale Land Surface Model. American Water Resources Association-Wisconsin Section Annual Meeting. Online. Mar 2021. *Oral.*
>> WI AWRA Outstanding Graduate Oral Presentation

Parsen, M., **C.B. Voter**, C. Hein[m], and A.H. Pruitt. Lake-Groundwater Interactions of Plainfield, Long, and Pleasant Lakes in the Central Sands of Wisconsin. American Water Resources Association-Wisconsin Section Annual Meeting. Online. Mar 2021. *Oral.*

earlier **Voter, C.B.** and S.P. Loheide II[m]. Advancing understanding of single family parcel hydrology to improve the hydroecologic outcomes of low impact practices. American Geophysical Union Fall Meeting. San Francisco, CA. Dec 2019. *eLightning.*
>> Invited

Voter, C.B. and S.P. Loheide II[m]. Soil amendment as a green infrastructure practice in residential areas. American Water Resources Association-Wisconsin Section Annual Meeting. Delavan, WI. Mar 2019. *Oral.*
>> WI AWRA Outstanding Graduate Oral Presentation

Nohr, K.[ug], **C.B. Voter** and S.P. Loheide II[m]. Comparing soil infiltration capacity among and within residential parcels in Milwaukee. American Water Resources Association-Wisconsin Section Annual Meeting. Delavan, WI. Mar 2019. *Poster.*

>> *WI AWRA Outstanding Undergraduate Presentation*

Voter, C.B. and S.P. Loheide II[m]. Quantifying weather-driven differences in the hydrologic outcomes of low-impact practices. American Geophysical Union Fall Meeting. Washington D.C. Dec 2018. *Oral.*

>> *AGU Outstanding Student Presentation Award*

Voter, C.B. and S.P. Loheide II[m]. Climate- and development-driven heterogeneity in hydrologic fluxes from urban residential parcels. Long Term Ecological Research Network – All-Scientists Meeting. Monterey Bay, CA. Oct 2018. *Poster.*

Voter, C.B. and S.P. Loheide II[m]. Introducing a real-world design problem to an intro water resources engineering course: effects on cognitive skills, confidence, and perception of value. UW-Madison Teaching and Learning Symposium. Madison, WI. May 2017. *Poster.*

Stieve, J.[ug], **C.B. Voter** and S.P. Loheide II[m]. Particle Size Analysis of Milwaukee Soils. UW-Madison Undergraduate Research Symposium. Madison, WI. Apr 2017. *Poster.*

Calderon, M. [ug], **C.B. Voter** and S.P. Loheide II[m]. Soil Particle Size Analysis in Milwaukee. UW-Madison Undergraduate Research Symposium. Madison, WI. Apr 2016. *Poster.*

Invited Seminars, Symposia and Panels (selected)

2023 Connecting Science, Management, and Community in the Central Sands and Beyond. Water@UW-Madison Symposium. Apr 2023. *Oral + Panel.*

2022 Urban ecohydrology: Processes that matter, ways to model them, and implications for sustainable management. Georgia State University. Nov 2022. *Oral.*

Scaling up: Accounting for key fine-scale urban hydrologic processes at city and larger scales. UMBC Center for Urban Environmental Research and Education Seminar. Online. Feb 2022. *Oral.*

What matters most? Tracking down key ecohydrologic processes in urban and agricultural landscapes to inform management strategies. University of Delaware Earth Sciences Seminar. Feb 2022. *Oral.*

2021 Using Co-productive Science to Address Wicked Water Problems in the Highly-irrigated Wisconsin Central Sands. University of Delaware Project WiCCED Seminar. Online. Oct 2021. *Oral.*

Advancing Science and Management through the Central Sands Lakes Study. UW-Madison Environmental Chemistry and Technology Seminar. Online. Oct 2021. *Oral.*

The Central Sands Lakes Study: Findings & Recommendations Overview. Wisconsin Department of Natural Resources. Online. May 2021. *Oral*. dnr.wisconsin.gov/topic/Wells/HighCap/CSLStudy.html

Tackling Wisconsin's Water Challenges through UW Water Science-Policy Fellowships and Agency Partnerships. *Panel*. At:

- Wisconsin Water Week, plenary session with panel discussion. Online. Mar 2021.

- University of Wisconsin-Madison 5th Annual Symposium for Research Administrators, panel discussion. Online. Sep 2020.

2020 Defining a "significant" reduction in "average seasonal lake levels" in the Wisconsin Central Sands. *Oral.* At:
- Wisconsin Potato and Vegetable Growers Association. Online. Mar 2020.
- Water@UW-Madison Spring Symposium. Online. May 2020.
- Wisconsin Lakes and Rivers Partnership. Online. Oct 2020.

2019 Wisconsin and Great Lakes Fellowships: Connecting Science, Policy, and Management. Water@UW-Madison Fall Poster Session. Madison, WI. Nov 2019. *Poster.*

Improving the quality of my research on urban residential hydrology by increasing the quantity of simulations using HTCondor. HTCondor Week. Madison, WI. May 2019. *Oral.*

Protecting our Urban Waters. Wisconsin Chapter of the Society for Conservation Biology, presentation and panel discussion. Madison, WI. Feb 2019. *Oral.*

earlier Delta Future Faculty Panel. Center for Integration of Research, Teaching, and Learning Fall In-Person Meeting, panel discussion. Madison, WI. Oct 2018. *Panel.*

Low-impact practices in residential areas: how interactions among practices and with climate alter urban hydrology. Marquette University Environmental and Water Resources Engineering Seminar. Milwaukee, WI. Oct 2018. *Oral.*

Low-impact practices on residential parcels: How and where they cause the biggest (hydrologic) bang. Nelson Institute Brown Bag Series. Madison, WI. Apr 2018. *Oral.*

Variability in deep drainage from urban residential parcels: what interventions have the greatest potential impact? Wisconsin Groundwater Association Speaker Series. Madison, WI. Sep 2017. *Oral.*

TEACHING

University of Delaware

CIEG 305 Fluid Mechanics, *3 credits*. Large, required course typically taken junior year by all a) civil engineering, b) environmental engineering, and c) construction & engineering management students. Introduction to the fundamental properties of fluids and characteristics of fluid flow, with a primary emphasis on solving engineering problems related to forces in a static fluid, conservation laws, dimensional analysis, and flow in pipes and open channels.

Enrollment:

2025 Fall	85 students (all undergraduate)
2024 Fall	97 students (all undergraduate)
2023 Fall	104 students (all undergraduate)
2022 Fall	109 students (all undergraduate)

GEOL 448/648 Ecohydrology, *3 credits*. Small- to medium-size elective that is dual-listed (undergraduate and graduate) and cross-listed (with Civil, Construction, and Environmental Engineering). Covers mutual interactions between the hydrologic cycle and ecosystems, with a dual focus on 1) developing and interpreting models that represent soil moisture dynamics, infiltration and overland flow, vadose zone processes, evapotranspiration, and groundwater dependent

ecosystems, and 2) presenting and discussing research articles from the current ecohydrology literature that relate to course topics. As of Spring 2026, coding is in python and assignments (optionally) incorporate guided use of AI tools in a manner that fosters critical thinking.

Enrollment:

2026 Spring	36 students (10 graduate / 26 undergraduate)
2024 Spring	28 students (8 graduate / 20 undergraduate)
2023 Spring	14 students (10 graduate / 4 undergraduate)

Guest & Workshop Teaching

2026 Fall	**CEE 311: Hydroscience**, UW-Madison *Topic*: Evapotranspiration
2022 Spring	**CIEG 403: Sustainable Applications in Infrastructure**, UD *Topic*: Green Stormwater Infrastructure
2017 - 2019	**Software and Data Carpentry Workshops**, UW-Madison *Topics*: Shell, Git, Intro to R *Instructor for (4) half-day units of a 2-day workshop for scientific researchers*
2019 Fall 2016 Fall	**CEE 311: Hydroscience**, UW-Madison *Topic*: Urban Hydrology
2014 Fall	**CEE 411: Open Channel Hydraulics**, UW-Madison *Topic*: Flow Routing

Professional Development (selected)

2023 **Course Design Institute,** Center for Teaching and Learning, University of Delaware

4-day workshop during which participants create student learning outcomes, a major assignment/exam to align with the new outcomes, learning activities that prepare students for the major assessment, and an inclusive syllabus.

2022 **Excellence in Civil Engineering Education (ExCEEd) Workshop,** American Society of Civil Engineers, Westpoint, NY

6-day workshop that provides engineering educators with theory, examples, practice, and feedback to improve their teaching abilities

2018 **Delta Certificate in Teaching, Research and Learning,** Delta Program, UW-Madison

Certificate awarded after successful defense of a teaching portfolio following science education coursework and a teaching-as-research internship. Committee: Dr. Trina McMahon (chair), Dr. Steve Loheide, Dr. Daniel Wright, Dr. Amber Smith, Dr. Rosemary Russ

2017 **Research Mentor Training**, Delta Program, UW-Madison

Semester-long seminar (1 credit) exploring strategies to become a more effective, culturally responsive mentor and discuss mentoring challenges as they arise

2016 **Delta Internship Program**, Delta Program, UW-Madison

Semester-long seminar (3 credits) in which each participant develops their own teaching-as-research project, implements it in a classroom, and presents findings

2015 **Teaching in the College Classroom**, Delta Program, UW-Madison

Semester-long course (3 credits) that provides a deep foundational knowledge of evidence-based teaching practices with opportunities to practice new teaching approaches

MENTORING

bold = current mentees

Postdoctoral Researchers

2025 -	**Dannielle Pratt**	*Co-advised with Holly Michael*	**UD**

Graduate Mentees

2026 -	**Mira Potter**	**PhD student**	**Civil Engineering**	**UW-Madison**
2024 -	**Emma Ruggiero**	**PhD student**	**Water Sci. & Policy**	**UD**
2023 -	**Austin Farnum**	**PhD student**	**Water Res. Engr.**	**UD**
2022 -	**Omowumi Erukubami**	**PhD candidate**	**Geological Sciences**	**UD**
2022 -	**Rachel Zobel**	**PhD candidate**	**Water Sci. & Policy**	**UD**
2024 – 2024	Yamini Medapati	MS (non-thesis)	Data Science	UD
2020 – 2025	Aaron Alexander	PhD	Civil Engineering	UW-Madison

Undergraduate Research Mentees

2026 -	**Emily Berger**	**Environmental Engineering**	**UD**
2025 -	**Catherine Hunt**	**Environmental Engineering**	**UD**
2025 -	Ryan Knott	Environmental Engineering	UD
2026	Kylie White	Environmental Engineering	UD
2025 – 2026	Hazel Mullen	Environmental Engineering	UD
2025 - 2026	Brayden Rochester	Environmental Engineering	UD
2025	Quinton Holley	Civil Engineering	UD
2024 – 2025	Nicole Mnych	Environmental Engineering	UD
2024 – 2025	Chloe Gerstenbacher	Environmental Engineering	UD
2024	Grace Hepler	Environmental Engineering	UD
2022 – 2023	Abigail Pierson	Environmental Engineering	UD
2018 – 2019	Katlyn Nohr	Geological Engineering	UW-Madison
2018	Aaron Canton	Civil and Environmental Engineering	UW-Madison
2017 – 2018	Jared Stieve	Civil and Environmental Engineering	UW-Madison
2016	Alex Bauch	Environmental Sciences	UW-Madison
2015 - 2016	Martin Calderon	Civil and Environmental Engineering	UW-Madison

Graduate Committees

2025	Dannielle Pratt	Ph.D.	Water Resources Engineering	UD
2025	K.J. Olsen	M.S.	Structural Engineering	UD
2023	Nana Abena Osei Tutu	M.S.	Water Resources Engineering	UD
2023	Lauren Donati	M.S.	Water Science & Policy	UD

SERVICE

University of Delaware

2025 – 2026	*Member*	**Admissions Committee**, Water Science and Policy Graduate Program

2025 – 2026	*Member*	**Executive Committee**, Water Science and Policy Graduate Program
2025 – 2026 2022 – 2023	*Member*	**Graduate Committee**, Dept. of Civil, Construction, and Environmental Engineering
2023 – 2025	*Member*	**Leadership Team**, Dept. of Civil, Construction, and Environmental Engineering
2023 – 2024	*Member*	**Faculty Search Committee**, Environmental Engineering
2022	*Member*	**Faculty Search Committee**, Coastal Geotechnical Engineering

Professional

2025 -	*Member*	**Ecohydrology Technical Committee**, Hydrology Section, American Geophysical Union
2022 – 2026	*Representative*	**Consortium for the Advancement of Hydrologic Sciences, Inc.**, University of Delaware representative
2023	*Session Organizer*	**Effective and Equitable Green Stormwater Infrastructure in the Chesapeake Bay**, Bay-wide Stormwater Partners Retreat, Chesapeake Stormwater Network
2023	*Panelist*	**Chesapeake Bay Science Technical Advisory Committee Workshop**, Using Carbon to Achieve Chesapeake Bay (and Watershed) Water Quality Goals and Climate Resiliency: The Science, Gaps, Implementation Activities and Opportunities.
2016 – 2022	*Founding Contributor*	**RePicture Engineering** *Contributions: support summer program, Role Model profile, review grant proposals and papers, beta test new features*

Reviewer Activities

Manuscript Reviewer for: Nature Water, Science of the Total Environment, Environmental Research Letters, Journal of Hydrology, Water Resources Research, Ecohydrology, Journal of the American Water Resources Association, Journal of Sustainable Water in the Built Environment, Journal of Hydrologic Engineering, Journal of Water Resources Engineering, Journal of Contemporary Water Research and Education

Proposal Reviewer for: NSF Hydrologic Sciences (5x ad hoc, 1x panel), NSF Strengthening American Infrastructure (1x panel), Washington Sea Grant (1x ad hoc), Department of Energy (1x panel)

MEDIA AND PUBLIC ENGAGEMENT

Media

Coastal Stormwater-Wastewater-Groundwater research

2025 *UDaily*. Data-driven tools development. [article]

Green Stormwater Infrastructure research

2021 *Green Bay Waterfront*. New video explores greening of Milwaukee to combat heat island and flooding. [article]

2021 *Wisconsin Sea Grant*. Can green infrastructure influence the weather in Milwaukee? [video]

2020 *UW-Madison College of Engineering*. Living on the Edge. [article]

2020 *UW-Madison College of Engineering*. A green scene: Engineers take a comprehensive approach to mitigating urban heat islands. [article]

2019 *EOS Research Spotlight*. Strategies to Improve Urban Hydrology. [article]

2017 *UW-Madison News*. A Data Tool for Homeowners to Make Rain Gardens More Effective. [article]

2012 *Wisconsin Water Resources Institute*. A Little from a Lot. [article]

Wisconsin Water Resources Science-Policy Fellowships

2019 *Wisconsin Water Resources Institute*. Voter Returns as Water Resources Science-Policy Fellow to Shed Light on Central Sands Lakes. [article]

2015 *Wisconsin Water Resources Institute*. Voter Elects to Become Debut WRI Fellow. [article]

Wisconsin Central Sands Lakes Study

2021 *The Mid-West Farm Report.* DNR Submits Central Sands Lakes Study Findings. [article]

2021 *Wisconsin Public Radio*. Report: Hundreds of High-Capacity Wells are Drawing Down Lake Levels in Central Sands Region. [article]

2021 *AP News.* Study: High-capacity wells draining central Wisconsin lakes. [article]

2021 *Wisconsin State Journal.* DNR study finds irrigation responsible for draining lakes. [article]

Additional media

2024 *UDaily*. Building a data-intensive research workforce for the Mid-Atlantic. [link]

Professional Development

2018 **Green Infrastructure and Stormwater Management**: A workshop to explore decision support tools to support community resiliency workshop. Sea Grant Extension, Milwaukee, WI. 3-day workshop.

2017 **Community-Engaged Research Institute**, Sea Grant Extension, Lansing, MI. 5-day workshop.